

BEGINNER REPRODUCIBILITY GUIDE

U.S. Electricity and Water Stress Monitor

Final RISE 2026 models, frozen artifacts, retraining workflow and offline GUI

What this guide lets a beginner do

Open the one-file offline GUI, run the included frozen model package immediately, or download the three model-ready CSVs and retrain all four final pipelines.

1. What is being reproduced

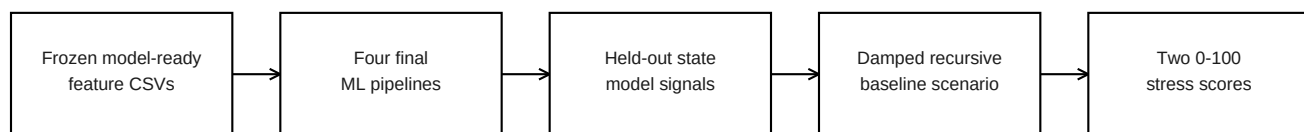
Pipeline	Prediction	Final model
SAIDI	Annual outage minutes/customer	Two-stage CatBoost
SAIFI	Annual interruptions/customer	Log-CatBoost
Drought	Drought severity, 0-100	ANOVA top 20 + Ridge
Compliance	Violation burden per 100,000 residents	CatBoost direct Tweedie

Electricity stress = 50% SAIDI stress + 50% SAIFI stress

Water stress = 50% drought severity + 50% compliance stress

Scope of exact reproduction

The final training script begins at the frozen model-ready CSV stage. A full raw-government-source rebuild additionally needs the exact source snapshots and separate data-engineering scripts.



Three supported user routes

OFFLINE GUI
download one HTML file

FROZEN MODELS
run the CLI immediately

RETRAIN
download CSVs and refit

2. Scientific scope and official sources

The framework was motivated by electricity unreliability and water insecurity in Pakistan. The modeling study uses granular U.S. public records as a test bed because consistent machine-readable local history was limited.

Official source	Role in the project
EIA Form EIA-861	Utility SAIDI/SAIFI targets, customers, ownership and reporting
Other EIA electricity data	Sales, prices, capacity, generation and demand context
NOAA climate and Storm Events	Precipitation, temperature, storm exposure and damage
Eagle-I	Recorded outage episode context
U.S. Drought Monitor	Weekly county D0-D4 drought percentages
U.S. Census and BEA	Population, poverty, income and GDP
EPA SDWIS	Public-water-system inventory and health-based violations
WRI Aqueduct	Static structural water-risk context

3. Why three tables create four models

Table	One row means	Unique key
Electricity	One annual utility-state target with 12 prior-year months	utility + state + target year
Drought	One county on one USDM week	county + map date
Compliance	One public water system in one year	PWSID + year

Compliance unit

Each system contributes health-based violations divided by state population and multiplied by 100,000. Summing system contributions reconstructs the state-year burden per 100,000 residents.

Frozen-input policy

The downloadable files are frozen model-ready tables. This prevents later revisions to government portals from silently changing the poster results. Input fingerprints are recorded in project metadata.

4. Cleaning, joining and leakage rules

- Read geographic and entity IDs as text so leading zeroes remain.
- Standardize state FIPS to two digits and county FIPS to five digits.
- Reduce context tables to their intended join key before merging.
- Validate joins as one-to-one or many-to-one and audit row counts.
- Sort by entity and time before creating lags or rolling histories.
- Shift before rolling so the current target cannot predict itself.
- Fit imputation, scaling and feature selection on training rows only.
- Exclude targets, target decompositions and same-period outcomes.
- Preserve chronological training, validation and backtest boundaries.

Why shift(1) matters

Without a prior-period shift, the current target can enter its own features. That is target leakage and makes evaluation look unrealistically strong.

5. Model-ready files

Filename	Approx. size	Training record
electricity_features_final_sequence_2013_2024.csv	74 MiB	Utility-state annual sequence
water_drought_features_county_week_2010_2024.csv	647 MiB	County-week
water_compliance_features_public_water_system_year_2010_2024.csv	835 MiB	System-year or residual

Main feature families

Pipeline	Examples
Electricity	Prior SAIDI/SAIFI; demand; storms; outage history; climate; capacity; utility categories; availability flags
Drought	Drought lags and rolling history; precipitation; temperature; calendar; socioeconomic and structural risk
Compliance	Prior violations; system inventory; population; socioeconomic; climate, drought, water-use and missing-data context

Large CSVs are hosted outside GitHub. The repository's downloader places the exact filenames in `Reproducibility/data/` and verifies them.

6. How each final model works

SAIDI - two-stage CatBoost

A classifier estimates the probability of a high-event year. Separate normal- and high-event CatBoost regressors are blended using that probability.

SAIFI - Log-CatBoost

CatBoost learns $\log_{1p}(\text{SAIFI})$. Predictions are converted back with $\expm1$, reducing the influence of extreme values.

Drought - ANOVA top 20 plus Ridge

Training-only median imputation and standardization are followed by ANOVA SelectKBest with the strongest 20 features and Ridge regression.

Compliance - CatBoost direct Tweedie

Direct Tweedie and two-part hurdle candidates compete on 2019-2020 validation data. Direct Tweedie wins and is used for the untouched backtest and deployment artifact.

7. Chronological evaluation policy

Stage	Years	Purpose
Selection training	Through 2018	Learn candidates
Validation	2019-2020	Choose settings and compliance family
Evaluation refit	Through 2020	Refit before backtest
Untouched backtest	2021-2024	Report final performance
Deployment refit	All observed	Save reusable artifacts

Out-of-sample history

2019-2020 signals come from models trained through 2018. 2021-2024 signals come from evaluation models refitted through 2020. Earlier predicted values remain null instead of being filled with in-sample fits.

8. Route A - open the offline GUI

At the repository root, download **Infrastructure_Stress_Monitor.html** and double-click it.

No installation required

The GUI works offline in a normal browser, requires no Python or server, and includes frozen observed values and baseline scenarios for all 50 states plus Washington, D.C., from 2025 through 2050.

The browser does not execute CatBoost or Ridge. It displays outputs prepared from the final frozen histories and scenario method.

9. Route B - use the included trained models

Windows PowerShell

```
git clone https://github.com/amessbee/rise26project.git
Set-Location ".\rise26project\Reproducibility"
py -3.13 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python stress_score_cli.py
```

No dataset download or model training is required for this route. Enter a U.S. state name or abbreviation and a four-digit year after 2024.

What is inside saved_models/

Artifact	Purpose
*.cbm and *.joblib	Frozen CatBoost and drought artifacts
state_model_history.csv	Observed and held-out model-signal history
project_metadata.json	Scenario method, boundaries and provenance
evaluation_metrics.json	Final validation and backtest metrics
feature_schema.json	Selected features and leakage audit
training_complete.json	Confirms a successful full run
SHA256SUMS.txt	Integrity hashes for the frozen package

10. Route C - retrain all four pipelines

Step 1: create and activate an environment

```
Set-Location ".\rise26project\Reproducibility"
py -3.13 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

Step 2: download the frozen model-ready CSVs

```
python csv_downloader.py
```

The downloader writes to data/, finds files nested by Google Drive, moves them to the data root and verifies every exact filename.

Step 3: deliberately replace the frozen model package

```
python train_models_once.py --force
python stress_score_cli.py
```

Resource warning

The drought and compliance tables contain millions of rows. Training can take a long time and needs substantial memory. Use a stronger computer or university server.

Optional: point to CSVs stored elsewhere

```
python train_models_once.py `
--electricity-file "X:\data\electricity_features_final_sequence_2013_2024.csv" `
--drought-file "X:\data\water_drought_features_county_week_2010_2024.csv" `
--compliance-file "X:\data\water_compliance_features_public_water_system_year_2010_2024.csv" `
--force
```

Without path arguments, the corrected training script automatically uses Reproducibility/data/. No author-specific drive path is required.

11. Final untouched state-year results

Target	Final model	MAE	RMSE	R2
SAIDI	Two-stage CatBoost blend	240.555	466.147	0.083
SAIFI	Log-CatBoost	0.219	0.305	0.660
Drought	ANOVA top 20 + Ridge	0.469	0.582	0.999
Compliance	CatBoost direct Tweedie	1.648	3.406	0.805

Interpret R2 carefully

R2 is not percentage accuracy. Drought's near-one result is a short-horizon, lag-rich held-out backtest. It is not evidence of nearly perfect distant-future forecasts.

12. How future scenarios are calculated

The CLI does not run trained models with imaginary future inputs. It:

- reads held-out state-year model-signal history;
- starts from the latest observed state value;
- takes the median change across the five most recent signals;
- advances one year at a time;
- halves the trend after every step;
- pulls each step 20% toward historical conditions.

Baseline scenario, not a direct forecast

Future values do not include known future rainfall, storms, population, demand, regulation or infrastructure. Approximate ranges use backtest RMSE and are not formally calibrated prediction intervals.

Software used for the final frozen run

Python	pandas	NumPy	scikit-learn	CatBoost	joblib
3.13.14	3.0.5	2.5.1	1.9.0	1.2.10	1.5.3

13. Final reproduction checklist

- requirements.txt installs gdown and the final model libraries.
- saved_models/training_complete.json exists for the frozen route.
- The CLI opens without a missing-package error.
- The downloader verifies all three model-ready CSV filenames.
- Retraining replaces saved_models only when --force is supplied.
- State names and abbreviations are accepted.
- Output includes all four targets and both stress scores.
- Compliance is labeled per 100,000 residents everywhere.

Common problems

Problem	Fix
gdown not found	Activate .venv and reinstall requirements.txt
Drive quota/access error	Retry later or request the frozen CSVs from the project owner
CSV not found	Run csv_downloader.py and keep exact filenames
Memory error	Close other programs or use a stronger server
Models already trained	Use them, or add --force to replace
CLI metadata missing	Restore the committed saved_models folder

Repository contents

```
rise26project/  
|-- Infrastructure_Stress_Monitor.html  
|-- README.md  
`-- Reproducibility/  
    |-- Beginner_Reproducibility_Guide.pdf  
    |-- Beginner_Reproducibility_Guide.md  
    |-- requirements.txt  
    |-- csv_downloader.py  
    |-- train_models_once.py  
    |-- stress_score_cli.py  
    `-- saved_models/
```

Reproduction complete

A beginner has reproduced the project when either the included frozen package or a successful retraining run lets the CLI return electricity and water baseline scenarios for a selected state and year.

Research-use warning

These scores support comparison and scenario exploration. They are not official utility, public-health or emergency-management forecasts and should not be the sole basis for operational decisions.